

Research Paper

PUTTING SOCIAL SCIENCES TO THE TEST

GROUP 5

AMNA SHAHID, MAHNOOR SHAIKH,
SYEDA RIDA FATIMA, MEHRAB JAN,
AFAQ KHAN

1. Introduction:

While primarily a global health crisis, the COVID-19 pandemic has led to a massive economic crisis, the effects of which are particularly concerning for low-income populations in less developed countries where the negative impact of this shock is not only more acute but is also more sustained. As such, it is important to investigate how households have adjusted to this shock in an environment of reduced household income, especially in the way resources and social investments are allocated within the household. Consequently, our paper will focus on how household decision-making about social investments has been impacted post-COVID-19. To understand our question better, we will explore whether there are differences in male and female preferences regarding social investments and how have social investments on children changed post the covid induced economic shock. The study will also look at how in the presence of an income shock, factors such as female agency and first and second-degree norms can cause heterogeneity with regards to social investments (i.e., household expenditure, children's enrollment, and women's labor force participation), allowing us to take into account both the quantitative and qualitative aspect of how and to whom resources are allocated within the household. We have also conducted a thorough literature review, particularly the works of DasGupta (differences in resource allocation), Khan (preferences for social investment), Field and Gulzar (first and second-order norms and women's self-perception and bargaining power), and Doss (the determinants of women's bargaining power) that acts as a framework for a better understanding of our outcome variables (differentials in parents' preferences, differentials in social investments on children and the impact of possible heterogeneities), and the resulting analysis. Our research and context are set in Chungi Amar Sadhu, a peri-urban area in Lahore, where we surveyed 173 households, following systematic sampling, to gain explanatory and exploratory data on how lives in low-income households have been affected post covid-19.

2. Literature Review:

Using the household as our unit of analysis, the unitary model predicts that households are altruistic and that distribution of income and assets within the household does not affect outcomes. However, Becker argues that even if different household members have different preferences, the household head is assumed to be altruistic. Yet, households may unexpectedly reallocate resources away from the transfer recipient. Hence, we consider the Intra-household Bargaining model, which suggests that different members have different preferences. However, the question arises, how do parents decide how much to invest in their children?

Understanding Household Income Loss and children's education:

Our primary motivation for conducting this research is to find how households have adjusted to economic shocks post-Covid and whether there are gendered differences to these adjustments. Jensen conducted a similar study, suggesting that households in areas impacted by adverse economic shocks (induced by rainfall shocks) may have a lower average income per capita and struggle to invest in the health and education of their children. Hence, children's social investments are reduced with unstable income and insufficient insurance opportunities. According to Jensen's analysis, compared to non-shock regions, children in the shock regions experienced a 20% point decline in enrollment (5), which is the same for girls and boys.

However, Dasgupta argues that "the impact of shocks on human capital attainment depends on how household-level income losses (or gains) translate into resource allocation within the household" (4). He argues that this impact can be especially acute for girls and women due to various factors, including gender-based discrimination and gender norms that may limit their access to education. Indeed, this change in household resources could affect early and later life investments in male and female children differently and becomes worth exploring, especially gauging attitudes and preferences for social investments.

Understanding Preferences towards Social Investments and the impact of female agency:

According to Sarah Khan, resource allocation becomes a function of preferences, which are, in turn, gendered since women are more inclined towards social investments than men (23). The paper talks about Pakistan's cultural and social norms that often restrict women's agency and autonomy and explores unexpressed preferences among women. The study found that, within married couples, women tend to have a greater preference for healthcare, drinking water, and income generation schemes and a lower preference for roads and transport than men (23). The study suggests that these differences in preferences may be due to the household's division of roles and labor (23) and other contextual factors, such as gendered norms of mobility and labor force participation rates (24).

Moreover, Doss states that women's bargaining power, the ability to influence the allocation of resources within a household, is important for women and can act as a measure of their empowerment because it can impact their ability to make decisions about the health and education of their children and promote their well-being. Since bargaining power cannot be directly observed, Doss measures it through proxies such as assets, income, education, and control over resources (1), the rationale being that women can contribute to spending decisions if they have control over their income (11). Education, in particular, maybe a helpful proxy because educated women tend to have different consumption patterns than illiterate women. It can be difficult to demonstrate empirically that women's bargaining power influences desirable outcomes, particularly for mothers and their children. However, data suggests that dynamics within the home impact household decisions. This study is relevant to our research since it provides a theoretical framework for understanding how unobservable variables, such as women's bargaining power, are shaped by household dynamics.

Understanding how first and second degree norms can shape women's bargaining power and labor force participation:

We have used Field and Gulzar's papers to delve deep into how norms affect women's self-perception, bargaining power, and preferences. Many women want to work; however, women who internalize gender norms governing workplace roles find that working outside the home dishonors their husbands (2343). Women's work is minimized through intra-household channels if the spouse internalizes these standards. By examining the effects of an exogenous rise in women's control over earned money, this paper seeks to understand how norms affect the labor supply of Indian women. Personal opinions and preferences, such as (i) whether or not women should be able to work outside the home and (ii) whether or not they wanted their sons to marry working women and to allow their daughters to marry a man who permitted work, were used to gauge norms (2351). Additionally, a vignette module featured two fictitious families from the respondent's caste who resided in the respondent's village (2368). In one situation, the wife had a paid job, whereas in the other, she stayed at home. Results show that women's bargaining power increases when money is deposited into their accounts rather than their husbands' accounts, which also lessens the influence of social norms adhered to by the husband. This article is pertinent to our study since researching the elements that affect women's decision to work is crucial, as working women will choose to invest in social causes differently than non-working women in lieu of their increased autonomy.

Lastly, Gulzar's paper focuses on individuals' beliefs about female political participation. The paper shows significant variation in people's expectations around patriarchal norms, with men and women likely to believe that others in their communities are less supportive of women's political participation than they actually are (2). This paper examines beliefs and expectations about specific behaviors and practices, particularly the distinction

between first and second-degree norms that influence what are deemed as appropriate behaviors for women, not just by other people but also by women themselves.

2. *Sample, Survey Design and Data:*

We employed a systematic random sampling of cross-sectional data in the region of Chungi Amar Sadhu by using the left-hand rule, where we surveyed every fourth house. A household was regarded as complete when it included both male and female household heads and at least one male and female child between the school-going ages of 3-21. The household size was defined according to the number of people who shared a '*choola*,' and the member being surveyed could only be considered the household head if they had decision-making power. The protocol instructed the female household head to be surveyed first, followed by the male head, to ensure that female answers were not anchored in male responses. Lastly, we followed randomization when our chosen household had multiple floors or households living in the same unit. The general distribution of our sampling is shown below:

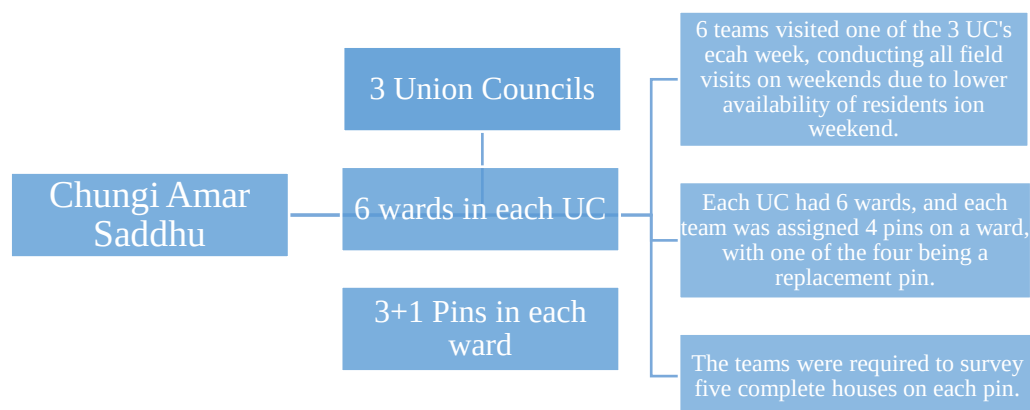
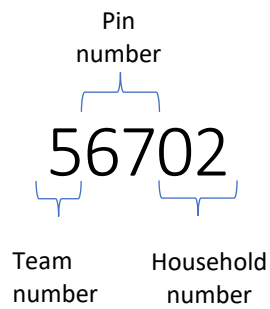


Fig1: Sampling Methodology

Our data was sorted at the household level, with each household having a unique identity, constructed in the following manner:



Our survey was designed on version 2.7 of the Survey CTO application and it consists of six modules as explained in the table below.

Module	Explanation
Introduction and Consent	This section filters out whether the household is relevant for the purpose of this study or not.
Household Head Attributes	The household head attributes capture the gender, age, education level, employment status and income of the household head.
Household Roster	For each household member, this section captured their gender, age, education level, and relation to the household head.
Employment and Income	This section gauged the total income (both Post and Pre Covid) of the household, the sector in which each member is working and if they received any help from the Govt or any NGO.
Intensity of shock	This segment captured the total current and expected consumption and expenditure of the household. It also inquires if any payment/expenses were deferred.
Decision making and Norms	This section asked questions about who makes the decision in the household. What do the respondents and the community think of women employment?
House Ownership and Assets	This section captured the attributes of the house, the ownership of the HH, and the assets owned by each member.
Network and Trust	This section gauged the social and familial network of the respondents and the strength of those networks.
Allocation of time	Here the survey captured how the respondent spends their day and how much time is allocated to different activities.
Preference assertion with regards to social investments	This section gauged the differences in the preferences of males and females by giving them hypothetical scenarios.
Fertility Choices	This section captured the fertility choices of the female respondents.

Fig 2: Modules of the Survey

4. Data and Context:

Our study was conducted in Chungi Amar Sadhu in November 2023, based on household-level surveys of male and female household heads in 173 households, facilitating intra-household comparison. Both male and female surveys include modules that capture data on income, expenditure, labor force participation, attitudes to social investments, and preference assertion. However, the female survey is more elaborate as it includes modules on norms, empowerment, decision-making, and bargaining power. This section describes the baseline summary statistics from the surveys we conducted. Table 1a represents a baseline summary of the description of the sample households and child enrollment in the household. Alongside this, it highlights intra-household gender roles, preference assertion, aspects of gender inequality within the household, and how it relates to labor market participation and expenditure choices.

In table 1a, February 2020 serves as a baseline to determine the conditions of each household before the shock. In our sample, the average household size was 6.5 people, with an average of 2.3 bedrooms in each house. The ownership of assets averaged 11.2 for all households. Pre-Covid 19, the mean household income in our sample was Rs.45,054 with an average expenditure of Rs.42,688. While the average monthly income increased to Rs. 48,046 in October 2022, it is important to note that soaring inflation has meant reduced purchasing power which is actually quite detrimental for poor populations. The change in social investments is measured through changes in expenditure on the health and education of children. For this purpose, the survey captured the child dropout rate. The proportion of salaried workers in February 2020 was 37% compared to 27% daily wage workers. This section allows us to compare the area's lived environment and income diversity. In our study, we believe that due to COVID-19, adverse economic conditions lead to an income shock in households, affecting the availability of household resources and leading to implications in children's social

investments. This change in household resources could affect early and later life investments in male and female children differently. It is evident in fig.6 that in 53% of households, at least one boy was enrolled in 2020 compared to 59% of households where at least one girl was enrolled pre-Covid.

A. Description of the sample households	Mean
Number of household members	6.55
Rooms in the house	2.35
HH asset count	11.22
Total monthly income in February 2020	45054.35
Monthly household expenditure in February 2020?	42688.89
Proportion of households where the household head is a daily laborer in 2020	0.27
proportion of households where the household head is a salaried worker in 2020	0.37

Table 1a.

C. Description of household heads	Male Respondents	Female Respondents	Difference
Labor force participation	0.809	0.405	0.404
years of formal education	6.97	5.2	1.77
Preference assertion	0.81	0.43	0.38
first order norms regarding female employment (held by)	0.62	0.36	0.26
second order norms regarding female employment (held by)	0.618	0.618	0

Table 1b.

Female labor force participation and education in our sample are significantly lower than the male counterpart, which is a crucial factor in women's bargaining power, as explored in the literature. We also observe that women are less likely to assert their preferences, thus reflecting intra-household power dynamics. Our study finds that women are less likely than men to make decisions independently about household purchases, visits to family, and pursuing employment or higher education, indicative of limited decision-making autonomy across

several domains. As illustrated in fig. 7, 42% of women make zero key decisions. Existing literature demonstrates how gendered norms constrain women's economic participation in Pakistan (Gulzar). We assess how this manifests in our sample households by examining first- and second-order norms regarding female employment. We find that more than 60% of men and 36% of women believe it is wrong for women to work outside their houses. Similarly, 62% of male and female respondents believe that most of society will censor working women (as shown in Table 1b).

Moreover, the questions in graphs 8a-8d serve as a proxy for differences in male and female preferences, which have essential implications for our study; for example, graphs 8a and 8b shed light on intra-household gender roles. In graph 8a, we observe that women prefer gas bills over petrol expenditure because they spend most of their time cooking for their family—a trend already hinted at by Sarah Khan in our literature review. Attitudes toward female labor force participation are further highlighted in graph 8b, where both men and women prefer work opportunities for male household members. Graph 8d shows a gender-specific question for children to test for any difference in preferences between investing in a girl and a boy child. It is evident that there is a greater preference for education opportunities of a female child during an income shock, maybe because boys are expected to earn in times of financial constraints.

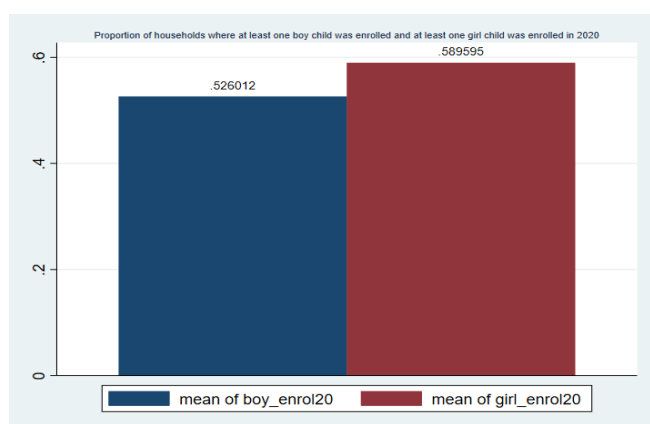


Fig. 6: boy vs girl enrolment in 2020

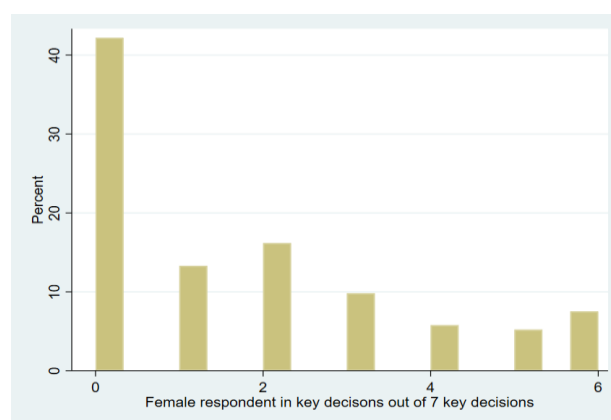
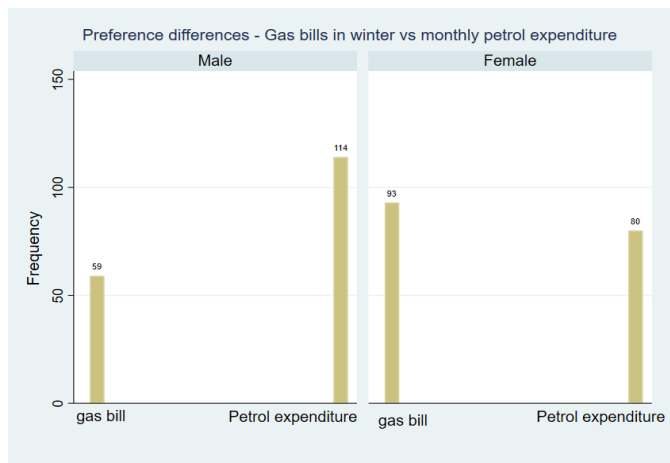
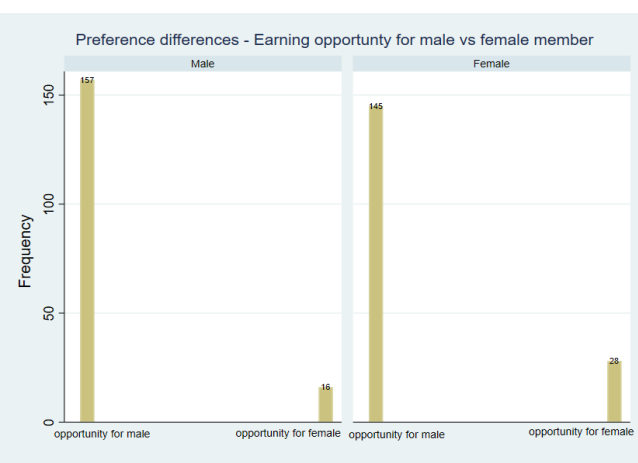


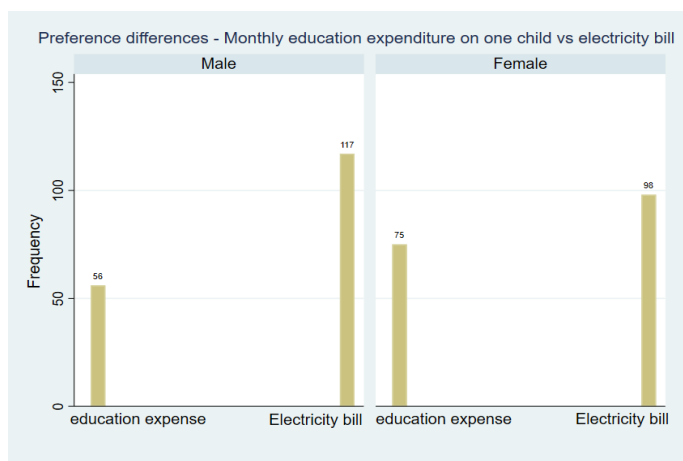
Fig. 7: Female decision making power



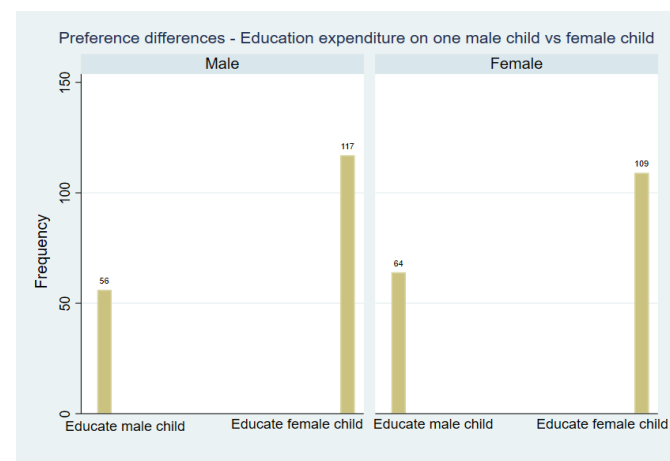
Graph 8a



Graph 8b



Graph 8c



Graph 8d

5. Hypotheses, Measures and Empirical Specifications:

This section focuses on the measures we have used to analyze how social investments have changed the covid induced income shock. We describe social investments as resources spent on children's education, household expenditure on food, other items, and personal expenditure. Based on our literature review (Khan and Doss) and understanding of the context above (as seen in Graphs 8 a, b, c, d in the Context section), we hypothesize that men and women have different preferences when it comes to social investments and so if women do in fact have greater agency and decision-making power, it can be expected that greater resources are spent on children's education and the overall well-being of the household (Sarah Khan). Our hypotheses are described as follows:

Hypothesis	Literature Referred to	Variables focused on
1. <i>In an environment of reduced income, households</i> a. <i>a)reduce expenditure on education</i> b. <i>b)allocate more resources to boys than girls (education and other expenses)</i>	1. Jensen 2. Sabarwal 3. Das Gupta	Effect of hh_loss_winc on: Male_mem_sch_out Fem_mem_sch_out Cutback_boy_exp Cutback_girl_exp

Table 2: Hypothesis summary

Since our research seeks to explore how households have adjusted post-income shocks, it is important to understand the intensity of this loss:

A) Household Intensity

Variable	Obs	Mean	Std. Dev.	Min	Max
dec_income	346	.4075145	.4920836	0	1
s2_q7_1 (risked not paying rent)	346	.2976879	.4579037	0	1
s2_q7_2 electricity bill	346	.7109827	.4539625	0	1
s2_q7_3 other utility bills	346	.4710983	.4998869	0	1
s2_q7_4 loans	346	.3988439	.4903697	0	1

Table 3: Intensity of HH Income loss

The intensity of the household shock is gauged by the overall decrease in income-on average, 41% of the households in Chungi have reported some kind of decrease in income. We further collapse this decrease into the households' inability to pay certain kinds of bills as a measure of the extent to which they were affected by the shock. The intensity ranges from payment of rent (the greatest amount in absolute terms) to phone bills (the least amount in absolute terms)-the intensity will be the greatest where households fail to pay even the meagre amount of phone bills (13.6%). It will be relatively lower where the maximum that they were unable to pay was their rent (29.7%). It can also be seen that the majority of these households were unable to pay electricity bills (71%), followed by other utility bills (47%), and lastly,

loans (40%). Following our surveys, these trends may be explained by the fact that maintaining a roof is much more important than having access to electricity and other utilities, especially during winter, which is why households may choose to cut back on this expense.

Index for hh loss	Freq.	Percent	Cum.
0	46	13.29	13.29
1	64	18.50	31.79
2	69	19.94	51.73
3	67	19.36	71.10
4	57	16.47	87.57
5	29	8.38	95.95
6	14	4.05	100.00
Total	346	100.00	

Fig. 3: Indexed Intensity of HH Income loss

We also take into account the index for household income loss that is created as the sum of all items that represent a loss incurred by the household. We see that 4% of our households had incurred losses across all dimensions whereas 86.7% of our sample households had incurred losses across at least one dimension.

B) The impact of household loss intensity on social investments:

1. For children

VARIABLES	(1) girl_dropout	(2) boy_dropout	(3) male_mem_sch_out	(4) female_mem_sch_out	(5) cut_back_boy_exp	(6) cut_back_girl_exp
hh_loss_winc	0.00804 (0.0193)	0.0335** (0.0139)	0.0609*** (0.0189)	0.0753*** (0.0157)	0.0797*** (0.0172)	0.0828*** (0.0203)
hh_rooms	-0.00775 (0.0280)	-0.0177 (0.0238)	0.0157 (0.0240)	0.0249 (0.0255)	-0.00494 (0.0254)	-0.000939 (0.0255)
hh_asset_count	0.00851** (0.00345)	0.00761** (0.00358)	-0.00731* (0.00421)	-0.00620* (0.00334)	-0.00139 (0.00437)	-0.000857 (0.00359)
resp_age	0.00139 (0.00189)	0.00493** (0.00229)	0.00297 (0.00200)	0.00311 (0.00218)	-0.00109 (0.00265)	0.000140 (0.00237)
resp_educ_years	0.00741 (0.00442)	-0.00120 (0.00403)	-0.00191 (0.00426)	-0.00463 (0.00448)	-0.00510 (0.00465)	-0.00182 (0.00465)
hhh_dailylabor20	0.00208 (0.0523)	0.0764 (0.0719)	0.0485 (0.0809)	0.0469 (0.0740)	-0.0237 (0.0460)	-0.00344 (0.0458)
fem_resp	0.0198 (0.0152)	0.0223 (0.0167)	0.0464 (0.0356)	0.0443 (0.0304)	-0.0129 (0.0421)	0.00401 (0.0455)

Table 4a: Regression to gauge changes in social investments (enrollment and expenditure) by gender wrt hh loss in income
**we have controlled for team fixed effects and clusters errors at the pin level

To explore whether enrollment rates have changed post covid for male and female children differently, we look at columns 3 and 4, which are highly statistically significant at 1%. We see that for every 1 unit increase in the intensity of loss in household income between

2020 and now, the dropout rates increase by 6% for boys and 8% for the girls-this difference is not as acute as we expect it to be. It seems that households value their children's education regardless of gender, even during a negative income shock. With regards to investigating whether a decrease in household income led to a cut in expenditures (other than education) for male and female children, we see that a 1 unit increase in the intensity of loss in household income increases the cuts in boys' expenditure by approximately 8%. In contrast, for female children, the cuts increased by 8.3%-again, a relatively small difference but highly significant nonetheless. This means that when encountering a tradeoff between children's education and cuts on their expenditure, parents in Chungi chose to prioritize their children's education, regardless of their gender-an optimistic result that allows us to reject our hypothesis (Table 2).

We also observe that the wealthier the household, the less likely that their children will drop out (3 and 4), i.e., for every one-unit increase in the household asset, the male and female dropout rate falls by 0.7% and 0.6%, respectively. Lastly, the table alludes to a plausible conclusion that the greater the education of the parent, the lesser the likelihood that they will compromise on their children's education-for instance, for every one-unit increase in parents' formal education, the dropout rate falls by 0.1% and 0.4% for boys and girls, respectively.

B: On other household expenses such as health and clothes:

	(7)	(8)	(9)	(10)
	exp_lower_self	exp_lower_clothes	exp_lower_health	cut_back_hh_exp
VARIABLES				
hh_loss_winc	0.0790*** (0.0144)	0.0874*** (0.0141)	0.0897*** (0.0184)	0.0494** (0.0202)
hh_rooms	0.0114 (0.0198)	0.00538 (0.0219)	-0.0100 (0.0220)	0.00348 (0.0185)
hh_asset_count	-0.00466** (0.00187)	-0.00492** (0.00243)	-0.00268 (0.00341)	-0.00540** (0.00210)
resp_age	0.000676 (0.00205)	0.00115 (0.00206)	9.41e-05 (0.00213)	-0.000234 (0.00191)
resp_educ_years	0.00778* (0.00408)	0.00864* (0.00439)	0.00639 (0.00542)	-0.00329 (0.00428)
hhh_dailylabor20	0.0119 (0.0399)	-0.0176 (0.0463)	0.124** (0.0478)	0.0103 (0.0421)
fem_resp	0.0688* (0.0366)	0.0619* (0.0366)	-0.0153 (0.0508)	0.0238 (0.0257)
-				

Table 4b: Regression to gauge changes in social investments wrt hh loss in income
 **we have controlled for team fixed effects and clusters errors at the pin level

Our regression analysis shows that for every one unit increase in the intensity of loss in household income, our respondents lowered expenditure on clothes and health by 8.7% and 9%. Contrastingly, the overall expenditure on the household decreased by 5%, which means that people preferred compromising on other types of social investments (health and clothes) compared to their children's education. Indeed, during our surveys, we often would get the impression that unless the respondents were undergoing a severe health issue, they would disregard the need to see a doctor. More often than not, they would self-prescribe.

stats	exp_lo~f	exp_lo~h	exp_lo~s
mean	.8612717	.5895954	.8150289

Fig 4: Decreases in self expenditure, health expenditure, and expenditure on clothes for women

stats	exp_lo~f	exp_lo~h	exp_lo~s
mean	.8208092	.6300578	.7861272

Fig 5: Decreases in self expenditure, health expenditure, and expenditure on clothes for men

Interestingly, although both male and female respondents have reported cuts in self-expenditure (86% for women and 82% for men), that are much greater than the cuts they have made on their children's expenditure, indicating a somewhat altruistic behavior from parents (Becker).

6. Heterogeneity Analysis and Discussion:

This part of the paper will investigate the impact of women's decision-making and first and second-degree norms on children's education (5&6), other expenditure on children (7,8), household expenditure (10,11,12), and lastly, women's labor force participation (1,13,14)

1. When women have greater agency and bargaining power (proxied by decision-making), children's education is prioritized and gendered differences in social investments between the boy and girl child are minimum, even during income shocks	1. Cheema 2. Sara Khan 3. Doss	Heterogeneity analysis of income loss and decision-making power-expenditure on education and other expenses for children
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2. When unfavorable first and second order norms exist, women are less prone to participating in paid labor and girl dropout rate will be higher than boy dropout rate, even during income shocks	1. Gulzar 2. Field 3. Doss	Heterogeneity analysis of income loss and first and second degree norms-flfp, girls dropout rate
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Table 5: Heterogeneity Analysis

	(1)	(5)	(6)	(7)	(8)	(10)	(11)	(12)	(13)	(14)
	Binary: Other women in the hh currently work	Boy(s): ever dropped out since 2020	Girl(s): ever dropped out since 2020	Delay or cut back expenditure on boys	Delay or cut back expenditure on girls	Respondent cuts back on personal clothing exp	Respondent cuts back on personal health exp	Reduced or delayed hh exp on food and improve ments	(Women) daily hours spent in paid work	(Women) daily hours spent working inside the home
Index for hh loss	0.040	0.049	0.061	0.092	0.105	0.094	0.078	0.058	0.162	-0.392
	(0.020)**	(0.025)*	(0.017)***	(0.022)***	(0.022)***	(0.020)***	(0.022)***	(0.029)*	(0.159)	(0.165)**
High female decision making power	0.039	-0.063	-0.008	0.183	0.159	-0.001	-0.105	0.088	0.012	-2.074
	(0.123)	(0.086)	(0.079)	(0.106)*	(0.120)	(0.099)	(0.105)	(0.106)	(0.627)	(0.756)***
hh_loss_me d_fem_dec_ inv	-0.080	0.025	0.028	-0.028	-0.048	-0.013	0.026	-0.019	-0.121	0.589
	(0.038)**	(0.031)	(0.027)	(0.027)	(0.031)	(0.028)	(0.032)	(0.030)	(0.234)	(0.269)**

Table 6.1: Heterogeneity based on women's decision-making power

**we have controlled for team fixed effects and clusters errors at the pin level

The results show that when women have greater than the sample median decision-making power, the number of working women in the household falls by 8% during an income shock. It is interesting to note that even though the coefficient on the household loss index is positive, implying that women are more likely to work when households undergo losses. However, in cases where women do have more agency (proxied by decision-making power), they are less likely to do paid work outside the home (falls by 12.1%) and work more inside the house by 59%. At first glance, this result seems contradictory to our expected hypothesis—women with a greater agency should ideally prefer paid work outside the home. However, the impact of negative first and second-order norms complemented by their own agency may mean that even though the house is experiencing a loss, if a woman could choose, she would not prefer to go outside the house to do work. Our survey findings complement this result. For instance, many women were running small businesses or shops inside their houses, allowing them more

flexibility in managing housework (which has increased post-income shock) and simultaneously earning for their families.

It is also evident that when women have greater decision-making power, the cutback on education (5 & 6) is negative, aligning with our hypothesis that the household prioritizes children's education (social investments) when women have greater authority to make decisions. However, it seems that the effect of household loss intensity outweighs the positive impact of female agency, so cutbacks on education increase during a shock even when women have greater agency.

Lastly, we notice that although an increase in household loss intensity expands the cutback on household expenses, when combined with female decision-making power, the magnitude of the cutback on household expenditures is lower, reflecting women's greater preference towards household well-being that can benefit all members of the household.

	(1)	(5)	(6)	(7)	(8)	(10)	(11)	(12)	(13)	(14)
	Binary: Other women in the hh currently work	Boy(s): ever dropped out since 2020	Girl(s): ever dropped out since 2020	Delay or cut back expenditure on boys	Delay or cut back expenditure on girls	Respondent cuts back on personal clothing exp	Respondent cuts back on personal health exp	Reduced or delayed hh exp on food and improvements	(Women) daily hours spent in paid work	(Women) daily hours spent working inside the home
Index for hh loss	-0.005 (0.024)	0.058 (0.023)**	0.063 (0.019)***	0.069 (0.027)**	0.068 (0.027)**	0.075 (0.017)***	0.109 (0.024)***	0.047 (0.023)*	0.363 (0.135)**	-0.345 (0.243)
Respondent : wrong for women to work	0.110 (0.123)	0.048 (0.067)	0.026 (0.062)	-0.010 (0.110)	-0.017 (0.112)	-0.155 (0.102)	0.059 (0.097)	0.005 (0.094)	1.946 (0.830)**	-1.346 (0.905)
hh_loss_wr ong_lfp	0.008 (0.036)	0.003 (0.028)	0.024 (0.024)	0.020 (0.033)	0.030 (0.032)	0.028 (0.025)	-0.039 (0.032)	0.005 (0.030)	-0.546 (0.290)*	0.461 (0.300)

Table 6.2: Heterogeneity based on first degree norms (wrong for women to work)

**we have controlled for team fixed effects and clusters errors at the pin level

According to first-order norms, as discussed in the literature, where both men and women believe that it is wrong for women to work outside the house, we see that hours spent in paid work reduce by 54.6%, whereas hours spent inside the home increase by 46% even in the presence of household income shock. These differences point towards how important first-degree norms can be when it comes to women's decisions about working in paid work-if both men and women feel that it is wrong for women to work outside the home, the household actively makes the choice of adhering to

this norm even if it comes at the cost of reduced income. This norm is also important in gauging the differentials in the social investments on education for children-we see that while the cuts in the dropout rate for both genders are the same due to intensity loss, they are higher for girls- when income loss interacts with first order norms-girls dropout by 2.4% whereas boys dropout by only 0.3%. It is reasonable to postulate that because girls will not be allowed to work, as deemed appropriate by the norm, it makes little sense to invest in their education, especially when household resources are already meagre because of the shock. Contrastingly, graph 8d in context illustrates that the likelihood of investing in a girl child is greater during an income loss maybe due to the perceived returns to education rather than norms.

	(1)	(5)	(6)	(7)	(8)	(10)	(11)	(12)	(13)	(14)
	Binary: Other women in the hh currently work	Boy(s): ever dropped out since 2020	Girl(s): ever dropped out since 2020	Delay or cut back expenditur e on boys	Delay or cut back expenditur e on girls	Responde nt cuts back on personal clothing exp	Responde nt cuts back on personal health exp	Reduced or delayed hh exp on food and improvement s	(Women) daily hours spent in paid work	(Women) daily hours spent working inside the home
Index for hh loss	-0.062 (0.034)*	0.051 (0.028)*	0.067 (0.027)**	0.112 (0.028)***	0.107 (0.028)***	0.089 (0.021)***	0.108 (0.029)***	0.069 (0.032)**	0.339 (0.261)	-0.053 (0.295)
Respondent : Society will censor working women	-0.174 (0.112)	-0.039 (0.080)	0.043 (0.077)	0.170 (0.108)	0.090 (0.109)	-0.041 (0.085)	0.040 (0.094)	0.149 (0.096)	2.317 (0.860)**	-0.663 (0.909)
hh_loss_so c_censor_lf p_maj	0.088 (0.039)**	0.014 (0.032)	0.008 (0.032)	-0.049 (0.034)	-0.035 (0.031)	-0.000 (0.022)	-0.025 (0.028)	-0.031 (0.030)	-0.398 (0.331)	-0.058 (0.341)

Table 6.3: Heterogeneity based on second degree norms (society will censor women who work)

**we have controlled for team fixed effects and clusters errors at the pin level

For second-order norms, we see that while household loss and the norm itself might decrease female labor force participation, the interaction coefficient is positive, indicating that the combined effect of these norms and household loss is not enough to keep women from work (an increase of 8.8%)- It could also mean that the women are not especially subservient to the second degree norms-during our surveys we came across a lot of women who believed that in times of financial distress, one should not let society restrain them (perhaps norms that dominate the household in a more intimate capacity have

a greater impact than those held by others). Nonetheless, second-order norms are crucial when it comes to paid household work, usually done outside the house. We see a decrease of 40% even when the impact of only household loss on women's labor force participation was positive. The impact of second-degree norms on children's education investment is interesting-we see that because of the norm, the differential in boys' and girls' dropout rate is substantial- the dropout rate for boys decreased (3.9%), whereas, for girls, it increased (4.3%)-much like the first-degree norms, albeit in a lesser magnitude, if parents think that society will sanction women who work, they might see little point in investing in their education.

7. Limitations and Analysis:

While our sample size is quite large, with 346 respondents, it is important to flag that despite maintaining all data collection protocols, the validity of our data was compromised due to various factors. Furthermore, while our regression analysis focuses on quantitative averages, the experience of conducting door-to-door surveys brought us face-to-face with many unique scenarios and outliers that can only be dissected through an anthropological lens. This section highlights some limitations of our quantitative data but also makes use of a case study to showcase how family dynamics and the presence of another family member can impact the answers obtained, exposing the façade of supposed agency in the process.

Firstly, the accuracy of our data was compromised due to recall bias in questions that required respondents to give answers off of memory about past income and expenditure, often times they would say "10-15,000 less than what we were earning" or simply, "we don't remember just write an arbitrary number." Apart from that, they were also influenced by their perception of our identities as students from an elite, liberal university, and we felt that specifically for questions about whether women should or not or whether their daughters would be allowed to work, they were cautious in terms of how we would perceive them and so would mostly answer in the affirmative.

In addition to this, oftentimes, the presence of another member affected the answers that women gave. While this was a recurring theme, albeit in a different context, in many surveys, it is in the following household case that this effect can be seen quite explicitly: We were conducting a survey

with a female household head of a nuclear family. At first, the woman was all alone when she was answering questions and did not hold back on mentioning how she struggles to make ends meet, and sometimes she has to cut back on her own expenditure so that their children can live a comfortable life. However, during the part of the survey where we explicitly asked whether she had to cut down on her expenditure, her husband had walked into the room to sit beside her, and he interjected, “No, we’re quite well to do”- and then the woman repeated his words verbatim. For many such answers, the man would hijack the survey and start answering on behalf of the woman, sometimes even pointing out how she was giving wrong answers and correcting them. This dynamic was quite interesting to see because, similar to this case, it prevailed even in cases where women reported having decision-making power which is our proxy for female agency. It remains unclear whether the man felt threatened in his role as the household provider when his wife felt she had to struggle to make ends meet or whether this behavior exhibited the power dynamic of their relationship where the man felt his voice was more important than his wife’s even when the question was addressed to her.

Conclusion:

Our research shows that the area of Chungi Amar Sadhu has experienced massive income losses post-Covid. While there are differentials in some preferences for social investments between men and women, these preferences remain similar on average for both girls and boys. It appears that in the event of a tradeoff between children’s education and other household social investments, such as expenditure on health or clothing, parents prioritize education. However, it is worth exploring why attitudes to female labor force participation (first and second-degree norms) are not particularly favorable in the region-the trends for female labor force participation indicate that both men and women prefer working inside the homes, even in the presence of an income shock. As such, any schemes or policies that aim to empower women should encourage formal female labor force participation and reduce the taboo around it. However, considering that variables such as norms are relatively hard to overcome, skill development programs can be initiated where women can work and earn from within the comfort of their households.

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